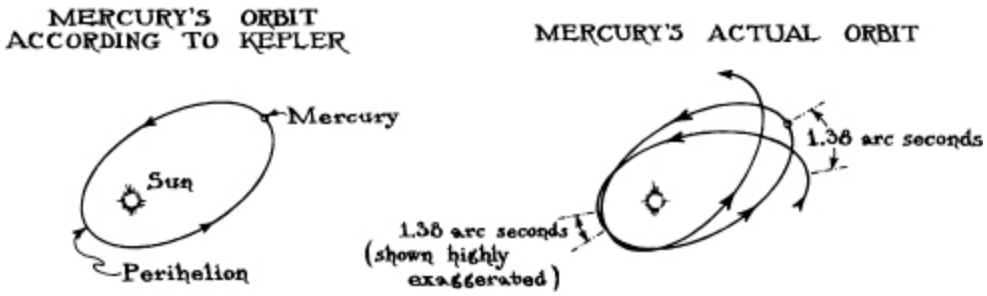
Kepler described the orbit of Mercury as an ellipse with the Sun at one focus (left diagram, in which the elliptical elongation of the orbit is exaggerated). However, by the late 1800s astronomers had deduced from their observations that Mercury's orbit is not quite elliptical. After each trip around its orbit, Mercury fails by a tiny amount to return to the same point as it started. This failure can be described as a shift, with each orbit, in the location of Mercury's closest point to the Sun (a shift of its *perihelion).* Astronomers measured a perihelion shift of 1.38 seconds of arc during each orbit (right diagram, in which the shift is exaggerated).

Newton's law of gravity could account for 1.28 arc seconds of this 1.38-arc-second shift: It was produced by the gravitational pull of Jupiter and the other planets on Mercury. However, there remained a 0.10-arc-second discrepancy: an *anomalous 0.10-arc-second shift of Mercury's perihelion during each orbit* The astronomers claimed that the errors and uncertainties in their measurement were only 0.01 arc second in size, but considering the tiny angles being measured (0.01 arc second is equivalent to the angle subtended by a human hair at a distance of 2 kilometers), it is not surprising that many physicists of the late nineteenth and early twentieth centuries were skeptical, and expected Newton's laws to triumph in the end.



Einstein's reasoning was simple: According to Newton, the gravitational force depends on the *distance* between the two gravitating objects (for example, the Sun and Mercury), but according to relativity, that distance is different in different reference frames. For example, Einstein's relativity laws predict that the distance between the Sun and Mercury will differ by about a part in a billion, depending on whether one is riding on Mercury's surface when measuring it or riding on the surface of the Sun. If both reference frames, Mercury's and the Sun's, are equally good in the eyes of the laws of physics, then which frame should be used to measure the